ASHA American Speech-Language-Hearing Association

AJSLP

Research Article

A Qualitative Analysis of Clinician Perspectives of Ultrasound Biofeedback for Speech Sound Disorders

Sarah Dugan,[a] Sarah M. Schwab,[a] Reneé Seward,[b] James Avant,[b] Ting Zhang,[b] Sarah R. Li,[c] Kathryn Eary,[c] T. Douglas Mast,[c] Michael A. Riley,[a] and Suzanne Boyce[d]

[a] Department of Rehabilitation, Exercise, & Nutrition Sciences, University of Cincinnati, OH [b] Myron E. Ullman, Jr. School of Design, University of Cincinnati, OH [c] Department of Biomedical Engineering, University of Cincinnati, OH [d] Department of Communication Sciences & Disorders, University of Cincinnati, OH

ARTICLE INFO

Article History:
Received June 14, 2022
Revision received October 14, 2022
Accepted January 14, 2023

Editor-in-Chief: Erinn H. Finke
Editor: Elaine Hitchcock

https://doi.org/10.1044/2023_AJSLP-22-00194

ABSTRACT

Purpose: Ultrasound biofeedback therapy (UBT) is a relatively new type of technology-assisted speech-language therapy and has shown promise in remediating speech sound disorders. However, there is a current lack of understanding of the barriers and benefits that may influence the usage behavior and clinical decision making for the implementation of UBT from a clinician perspective. In this qualitative study, we explore the perspectives of speech-language pathologists (SLPs) who have used ultrasound biofeedback in programs of speech sound therapy using the unified theory of acceptance and use of technology (UTAUT) model.

Method: Seven SLPs who had clinical experience treating speech sound disorders with UBT participated. Semistructured in-depth interviews were conducted and video-recorded. Two coders coded and categorized the transcribed data, with consensus established with a third coder. Using thematic analysis, the data were exploratorily grouped into themes along components of the UTAUT model.

Results: The highest number of codes was sorted into the "effort expectancy" theme, followed by "performance expectancy," "social influence," and "facilitating conditions" themes of the UTAUT model. Clinicians identified multiple perceived barriers and benefits to the use of ultrasound technology. The top identified barrier was limited accessibility, and the top benefit was the ability to visualize a client's articulatory response to cues on a display.

Conclusions: Clinicians prioritized "effort expectancy" and "performance expectancy" when reflecting on the use of ultrasound biofeedback for speech sound disorders. Clinicians spoke favorably about using UBT for speech sound disorder treatment but acknowledged institutional barriers and limitations at organizational and social levels.

Technology-assisted speech-language therapy (TASLT; Kuschmann et al., 2021) is a rapidly growing area of clinical practice in speech-language pathology. TASLT refers to technological support for face-to-face interaction and home practice (though not telehealth), including speech recording and analysis software and devices, as well as mobile applications known as mHealth. TASLT, as the name suggests, is a category specifically for therapy, not diagnostics. Many types of TASLT have been used to support speech and language therapy practice, as described in the following systematic reviews or illustrative examples: (a) electropalatography, which uses an artificial electronic palate to track tongue contact patterns (Hardcastle & Gibbon, 1997); (b) speech acoustic biofeedback, which uses graphic representations of therapeutic changes in the acoustic signal (e.g., McAllister Byun & Campbell, 2016); (c) electromagnetic articulometry, which tracks the

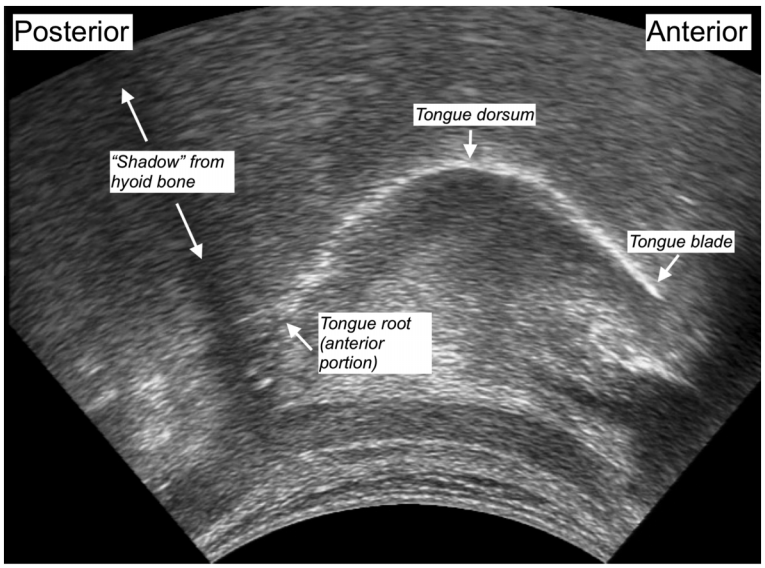
Correspondence to Sarah Dugan: hamilsm@ucmail.uc.edu. ***Disclosure:*** *The authors have declared that no competing financial or nonfinancial interests existed at the time of publication.*

 Copyright © 2023 American Speech-Language-Hearing Association

movement of sensors glued to articulators in a magnetic field (e.g., Katz & Mehta, 2015); and (d) ultrasound biofeedback therapy (UBT), which uses ultrasound imaging to display real-time movements of the tongue surface (Sugden et al., 2019).

UBT is one TASLT method that is rapidly gaining traction in clinical settings. Ultrasound biofeedback involves an ultrasound transducer array or probe placed under a client's chin to visualize the tongue surface and its real-time movement. Because ultrasound waves do not substantially penetrate interfaces between soft tissue and air or bone, most other articulatory structures, such as the teeth, lips, hard and soft palate, and nasopharyngeal port, are not visible on ultrasound images of speech production (see Figure 1). Depending on the location of the ultrasound transducer relative to the tongue and the sound being produced, portions of the tongue tip may be obscured due to acoustic shadowing caused by reflection of ultrasound from the sublingual tissue–air interface and the mandible bone. Similarly, portions of the tongue base may be obscured due to acoustic shadowing from the hyoid bone. The tongue surface from blade to dorsum, considered the most relevant and active portion of the tongue for speech, is reliably imaged. Although imaging is restricted to the tongue surface in the oral tract, multiple studies have indicated that motor-based therapy approaches incorporating ultrasound are effective for speech sound acquisition in children with speech sound disorders (Cleland et al., 2015; Modha et al., 2008; Preston, Leece, & Maas, 2017; Preston, McAllister, et al., 2018), children with apraxia (Preston et al., 2013, Preston, Leece, McNamara, & Maas, 2017), and children who are hard of hearing (Bacsfalvi, 2010; B. Bernhardt et al., 2003). The focus of UBT has often been on persistent difficulties with North American English /r/ (Boyce, 2015; Boyce & Lee, 2016; Dugan et al., 2019; Gick et al., 2008; Preston, McAllister Byun, et al., 2017). Ultrasound imaging for speech therapy was once largely restricted to university or hospital laboratory settings due to the high cost and relative bulk of ultrasound scanners but can now be found in an increasing number of university and hospital clinics, as well as a few private practices. Increased adoption of ultrasound has been made possible by the decreasing cost of clinical scanners and the increasing availability of portable, low-cost, smartphone-integrated transducers (Baribeau et al., 2020; Liu et al., 2019). For example, whereas cart-based ultrasound machines could cost upward of $100,000, new handheld and hand-carried ultrasound options integrate with commercial tablet or smartphone devices and are in the range of $2,000–$7,000 USD (Falkowski et al., 2020). Low-cost handheld ultrasound scanning devices have even been suggested for clinical field environments and potentially for use by patients themselves during telepractice (Clevert et al., 2019).

Figure 1. Ultrasound image of sagittal tongue (right anterior orientation) of tongue from adult female at rest, using Siemens Acuson X300 Premium Edition diagnostic ultrasound system with a C6-2 curved linear array transducer, image depth of 8 cm, center frequency of 4.0 MHz, and frame rate of 36 fps. The tongue–air interface corresponding to tongue surface is shown as a bright white curvilinear shape. As is typical with ultrasound views of speech, other articulatory structures, such as palate, velum, alveolar ridge, teeth, lips, and tongue tip, are not visible. Identified tongue regions (blade, dorsum, and root) are approximate and correspond with typical labels used by speech-language pathologists when cueing during therapy.



Most studies on UBT have concentrated on its efficacy for treatment of certain clinical populations, with most focused on careful control of treatment protocols and recruitment criteria, with success measured in terms of percent accurate productions of a targeted speech sound (Cleland et al., 2019; McAllister Byun et al., 2014; Preston, Leece, & Maas, 2017; Preston, Leece, McNamara, & Maas, 2017; Preston, McAllister, et al., 2018; Roxburgh et al., 2015; Sugden et al., 2019). Clinicians often wonder which children are most suitable for UBT, and some studies have included children's experience with UBT (Adler-Bock et al., 2007; Bacsfalvi, 2007; Preston, Holliman-Lopez, & Leece, 2018). Few, however, have focused on the clinician experience. The increasing use of UBT in university and hospital clinics suggests that practicing speech-language pathologists (SLPs) find value in the use of ultrasound during speech sound therapy, but the specifics of their opinions on the technology, their reflections on how the technology might be improved, and any need for further information or training are largely unknown. Perhaps the most informative discussion of consultative use of UBT comes from the study of M. B. Bernhardt et al. (2008), in which rural SLPs observed an ultrasound examination of their clients' speech and then used the information to treat their clients back in the field. In that study, SLPs were quoted as having positive impressions of the tool's overall effectiveness and noted that they learned new information about articulation and effective cue sequences from the experience. However, SLPs' experiences were not the primary focus of the study, and their experiences were not systematically analyzed.

Beyond the specifics of UBT as a useful technology, however, an inquiry into how clinicians think about UBT can be used to understand how SLPs integrate new TASLT modalities in their practices and provide insight into the process by which clinicians make diagnostic and treatment decisions. There are significant constraints on a clinician when deciding to implement new evidence-based practices: resources required, time involved, and access to information (Brackenbury et al., 2008; Diepeveen et al., 2020). In the case of ultrasound, the technology represents a significant monetary investment for a clinic, and using it relies heavily on clinical training and on clinical reasoning and interpretation. There are also factors that clinicians perceive as limitations to their decision-making abilities (Diepeveen et al., 2020). Many clinicians have been trained to work under a linear or formulaic model of clinical reasoning involving differential diagnosis and classification (McAllister & Rose, 2000), where clinicians focus on medical signs and symptoms to form a tentative hypothesis regarding diagnosis (Schwartz & Elstein, 2008). Integrating a new technology into this model is challenging because many of the elements clinicians have been trained to focus on are still in flux—for example, a new technology may reveal or emphasize existing discrepancies in diagnostic criteria, terminology, and clinical consensus (Diepeveen et al., 2020). Perhaps in response to these limitations, SLPs using a new technology often rely on their own prior experiences and on their social networks when making clinical decisions (Zipoli et al., 2005).

The purpose of this study was to elicit and interpret clinician perspectives on ultrasound biofeedback, with specific attention to barriers and benefits of technology usage and decision making when remediating speech sound disorders. As such, the study follows in the tradition of usability research, also known as "user testing" research, that aims to know how users from different groups think about and interact with a new technology. Because the field of speech-language pathology consists mainly of White, non-Hispanic women (American Speech-Language-Hearing Association [ASHA], 2020), we prioritized outreach to SLPs who have diverse demographic backgrounds for this study. The study takes a qualitative approach, on the assumption that qualitative methodology can elicit and interpret the experiences of clinicians in a systematic and valid way, providing a framework for later hypothesis-driven quantitative research. Qualitative methods are particularly well suited for addressing pervasive concerns in rehabilitation related to clinical reasoning and decision making, clinician–client interactions, social issues of health delivery, and explanatory mechanisms behind the harms or benefits of an intervention (VanderKaay et al., 2018).

The results of this study can be examined for correspondence to well-established models of technology adoption and decision making, such as the theory of reasoned action (TRA; Ajzen & Fishbein, 1980; Fishbein & Ajzen, 1975) and, more recently, the unified theory of acceptance and use of technology (UTAUT) model (Venkatesh et al., 2003). UTAUT is a popular behavioral model following from models such as the TRA, which posits that individuals behave in certain ways based on attitudes and intentions (Ajzen & Fishbein, 1980; Fishbein & Ajzen, 1975). The UTAUT model has been used to examine personal factors (i.e., background life and lifestyle of an individual) that may influence clinicians' use of various clinical technologies, such as clinical decision support systems (Jeng & Tzeng, 2012), mobile electronic records systems (Hennington & Janz, 2007; Kim et al., 2016; Wills et al., 2008), precision medicine technologies (Wang et al., 2020), and telemedicine equipment (Kohnke et al., 2014), among many others. By situating the qualitative data within known factors influencing technology use, the relative prominence of these factors can be determined for SLPs, providing a framework for future mixed-methods, hypothesis-driven research on technology adoption in speech-language pathology.

Method

The selection of methodological details reported in this section was guided by and adherent to the Consolidated Criteria for Reporting Qualitative Research checklist for interviews and focus groups (Tong et al., 2007). The study was approved by the University of Cincinnati Institutional Review Board.

Participants

Seven clinicians participated in the study. They were identified by the first author of this study and were either contacted privately via social media or e-mailed. Most participants were known due to shared interest in UBT or through prior clinical or academic affiliations. A follow-up e-mail was sent explaining the study for those contacted on social media who showed interest. No participants dropped out from the study once enrolled. Our sample size was planned based on findings in the user test literature, which indicate that a sample size of 5 is typically sufficient to provide information about factors impacting the usability of a technology (Nielsen & Landauer, 1993).

All participants had at least 15 hr of treatment or clinical research experience with ultrasound for speech sound disorders. Participants all had their clinical certificate of competency and had been practicing as SLPs for at least 2 years. All participants had experience with ultrasound as used clinically with speech sound disorders in children, which echoes the prevalence of this population in the research literature (Sugden et al., 2019). Most of the participants' research or clinical practice focused primarily on children, but one participant specialized in voice disorders across the life span and one was currently working in a rehabilitation facility for adults.

Because SLPs are overwhelmingly likely to be White, to be female, and to come from middle-class backgrounds (ASHA, 2020), there is a strong risk of recruiting a homogenous group of participants with similar backgrounds. Instead, we used purposive sampling (Palinkas et al., 2015) to recruit clinicians with differing life experiences and personal characteristics (e.g., race or gender), embracing the complexities and nuanced accounts that different life experiences can bring to a clinician–client encounter. As Table 1 below shows, we prioritized outreach to clinicians from diverse racial and ethnic groups.

Data Collection

Data were collected in one-on-one, in-depth semi-structured interviews over Zoom (Version 5.8.0). Participants chose the time and setting (e.g., home or clinic) most convenient for them. Participants were provided the interview questions at least 24 hr before their interviews via e-mail. Interviews were private; no nonparticipants were present during the interviews. All interviews lasted between 45 min to 1 hr and occurred during the last 2 weeks of October 2021. Participants voluntarily provided demographic data using an online form outside the interview setting (see Table 1). Participants were also asked in an e-mail to volunteer information about the dialects and languages they spoke outside of Standard American English.

Questions for the interviews were developed in an interdisciplinary way, using the different perspectives afforded by the interdisciplinary research team (i.e., SLPs, physical therapist, movement scientists, biomedical engineers, and communication designers). Our rationale was twofold. First, as innovation frequently is found at the boundary of disciplines (West, 2016), research suggests that interdisciplinary study is valuable in promoting unconventional thinking (Newell, 1994). Second, prejudices and biases specific to one particular discipline may be avoided when multiple disciplines are represented. Interdisciplinarity is particularly important for the study of UBT use among clinicians, as there is a need to understand multiple dimensions of usage behavior: clinical demands (SLPs), technological implementation (engineers), user

Table 1. Demographic information from participant sample.

Sex	*n*	Race	*n*	Ethnicity	*n*	Linguistic information	*n*
Female	6	African American	1	Hispanic	1	Standard American English and African American English, regional Southern influences	1
Male	1	Asian/Pacific Islander	1	Not Hispanic	6	Farsi	1
Not reported	0	White	3	Not reported	0	Japanese	1
		Other	2			Spanish with different Spanish dialects	1
		Not reported	0			No other languages other than Standard American English, no other dialects	1
						Not reported	2

Note. The total number of participants who reported demographic information for each category is reported to the right of the category.

experience and user interface (designers), motor control (SLPs, physical therapists, and movement scientists), and rehabilitation theory (SLPs and physical therapists). The members of our interdisciplinary team were familiar to one another from prior work, all with a history of conducting UBT research together for 2–10 years.

Interview questions were broadly aimed at identifying perceived barriers and benefits to using ultrasound for speech sound disorder treatment and diagnosis. There were four major parts to the interviews: (a) general background, (b) "pain points" (e.g., barriers), (c) benefits, and (d) person- and family-centered experience (see Appendix A). For each major part of the interview, discussion began with a broad, open-ended question. More specific follow-up questions were asked related to the opening questions to further probe clinician responses. All questions were reviewed and agreed upon by the authors of this article prior to study start. Video recording was used to collect data during Zoom sessions, and the interviewer made notes related to participant responses during the interview as points to further discuss. These notes were not analyzed, as all of the content in the notes was also present in the video recordings.

The interviewer was the second author. The interviewer was chosen to lead the interviews due to previous interview and qualitative experience and her work in a different allied health field (physical therapy), which allowed her to understand the general demands of speech-language pathology and rehabilitation while also being unknown to any participants. This last point was considered to be particularly important, as the field of ultrasound researchers and clinicians may be characterized as a socially dense network, with most people knowing each other or having multiple colleagues in common. To help each participant feel comfortable when discussing their experiences, a person unknown to them who shared general clinical experiences was desirable. The interviewer had over 2 years of dedicated research experience working with the interdisciplinary team on UBT projects with specific contributions related to her expertise in motor control and motor learning, qualitative methods, and critical rehabilitation and critical disability. Thus, the interviewer was deemed by the team as highly qualified to conduct the interviews and provide a unique perspective in "why clinicians do what they do." There were limits to the interviewer not being an SLP (e.g., more limited insight into nuances of SLP practice); however, the benefits outweighed the limitations. As we will further describe, an SLP was prominently involved in data coding and interpretation, minimizing the possibility for misinterpretation of findings because of disciplinary background. Before the interviews, participants were made aware of the interviewer's name, basic professional qualifications, and background (including specific contributions to this study).

Questions were asked during the interview in a consistent order, with follow-up questions asked when appropriate. Participants were provided the opportunity to e-mail the first or second author to bring forth additional questions or discussion points after the interview was over; however, no participant did this.

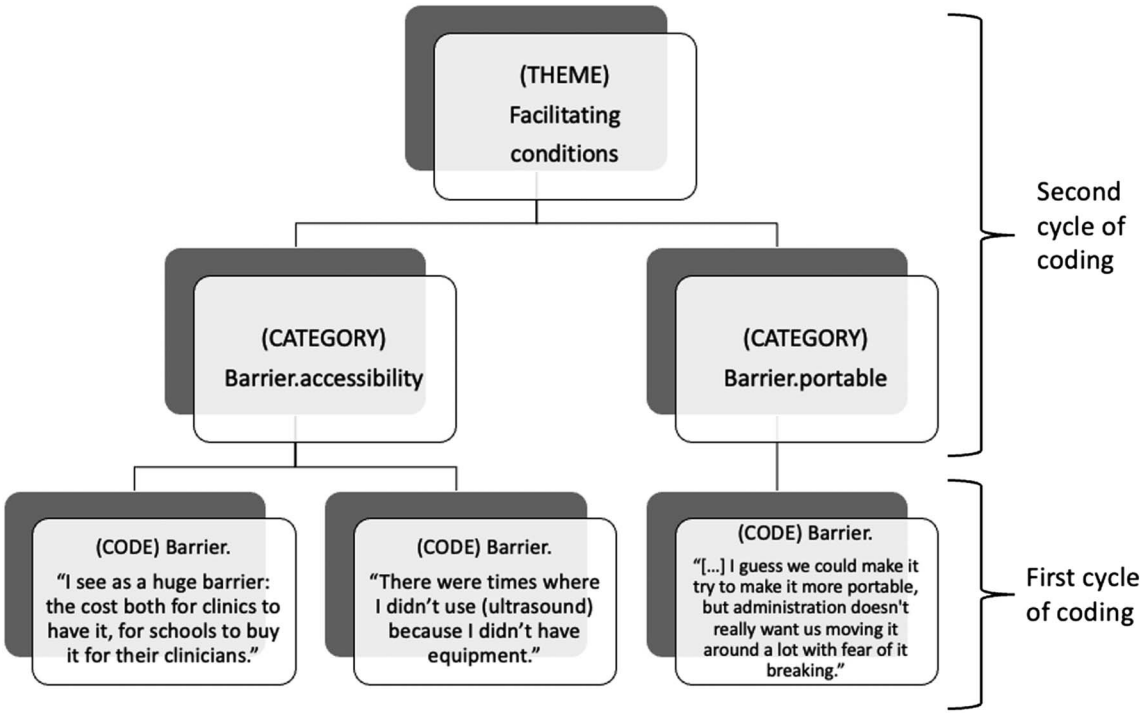
Data Analysis

After the interviews concluded, the second author transcribed the interviews from the Zoom video recordings verbatim into the open-source qualitative analysis software Taguette (Rampin, 2021) and coded the data, creating a coding tree. The first and third authors also coded data during the first cycle of coding (see Figure 2).

A qualitative methodology was employed in which codes were derived from the data and were not set a priori (Smith, 2004). Codes are tags that assign units of meaning to words, phrases, or sentences with reference to relevant phenomena in the data, allowing for the construction of a conceptual scheme that suits the data (Basit, 2003). Because the aim of these interviews was to determine what helped or hindered SLPs' adoption of UBT, the first coder transcribed the interviews and coded statements that referred to factors relevant to UBT and its use. This list of codes was produced using an iterative process, as the recordings were listened to and transcripts were read repeatedly by the first three authors (including an SLP) to identify recurring ideas within these general references, and then these were formally coded with descriptive titles that referred specifically to the meaning in each statement. For example, "barrier.cost for client" and "barrier.age" both refer to a barrier to UBT, but the statements referring to these codes were different and therefore required distinct codes in order to make meaningful sense of the data structure. Note that a statement may have had multiple codes assigned (e.g., a statement could reflect both cost and accessibility barriers), but the codes themselves referred to distinct concepts; otherwise, they would be collapsed during the coding review process. A coding manual was created to ensure consistency and validity of analysis (see Appendix B).

During coding review, after one person coded the data, the next coder reviewed the work and identified discrepancies. Percent agreement across coders was determined by counting the number of instances where coders disagreed divided by the total number of codes. There was high agreement across coders after the first pass of coding, with disagreement occurring on seven instances of 510 coded statements, for an overall agreement exceeding 98%. The high agreement is reflective of coders' excellent adherence to the coding manual and similar interpretation of results, and high agreement should not be

Figure 2. The process of qualitative analysis used for the interview data.



misinterpreted as an artifact of common mentorship or training as all coders completed graduate training under the guidance of different mentors and in different disciplines (speech-language pathology, physical therapy, and design).

When consensus was achieved, the three coders discussed data saturation. Data saturation refers to the point where information obtained in interviews can be seen as repeating across participants, and that additional participants would likely not contribute additional meaningful information beyond that of the current sample. As noted previously, data saturation in user test research is commonly reached with five participants. For this study, when consensus among coders was achieved, the three coders discussed data saturation, considering whether there was a detailed picture of "why clinicians do what they do" (Popay et al., 1998) with UBT. Total numbers of statements tagged with each code were counted. Recurrence was found in responses across the major interview questions. Based on the amount of information, as well as its consistency across individuals, it was determined that our study with seven interviewees had reached the data saturation point, and additional participants were not required (Saunders et al., 2018).

Codes were then grouped by the three coders into seven broad categories (see Appendix B for categories and codes in each) according to recurring patterns: (a) barriers: responses related to shortcomings or perceived negative features of ultrasound for speech sound therapy; (b) benefits: responses related to perceived positive features of ultrasound for speech sound therapy; (c) wants: responses related to desired features, uses, or what ultrasound might accomplish with modifications; (d) progress: responses related to factors that might limit or enhance progress in clients; (e) experience: responses related to clinician background; (f) concerns: responses related to family or child concerns or hesitancies; (g) success: responses related to how children define success in therapy; and (h) miscellaneous: responses related to specific questions aimed at eliciting a single answer, not reflection.

To identify thematic relationships among the data (e.g., shared views on the usage of technology), codes within categories were regrouped into four aspects of the UTAUT model (Venkatesh et al., 2003). The UTAUT model has received extensive empirical support for its ability to predict technology acceptance and adoption (e.g., Chan et al., 2010; Cheng et al., 2011; Yang, 2010; Zhou, Lu, & Wang, 2010). This model explains 70% of the variance in user intentions to use technologies, and it has been shown to outperform previous models (Venkatesh et al., 2003). The UTAUT model aims to understand and explain a person's intention to use a technology, as well as a person's subsequent usage behavior. As the objective of this study was to elicit and interpret clinician perspectives of ultrasound

biofeedback, with specific attention to barriers and benefits of using the technology that may influence usage behavior and subsequent clinical decision making, the UTAUT model was deemed relevant to achieving this objective.

The three direct determinants of a person's intention to use a technology according to the UTAUT are performance expectancy (the degree to which a person expects a technology to assist them in their job), effort expectancy (the ease of use associated with a technology), and social influence (the degree to which a person believes high-status people think they should use the technology). These three components are the direct determinants of behavioral intention to use the technology, and they, along with facilitating conditions (the degree to which a person believes that there is structural and organizational support for the technology), determine a person's usage behavior. Adopting the UTAUT model to provide names for high-level themes in the data was determined by the study team to provide a meaningful, interpretive structure that can be used in future investigations of clinician technology acceptance, inclusive of mixed-methods or quantitative approaches.

For this thematic analysis, the three coders collaboratively regrouped the identified codes and categories into the four components of the UTAUT model, which served as the themes for the overall qualitative study. The total number of codes and categories in each UTAUT component was counted and displayed in a revised coding tree to examine relationships among the items. Codes and categories were thematically grouped until consensus on all items was achieved. Information within and across themes were examined to encapsulate clinician perspectives on barriers and benefits to using ultrasound for speech treatment.

Results

This section first presents information and characteristics about the study participants obtained through the interview process. These are results that further inform about the demographics of the interviewed clinicians and are useful for understanding the unique backgrounds of the participants as a group. The results of our thematic analysis with categories and codes linked to the UTAUT model are then presented. We explored how clinician responses fit with each component of the UTAUT model, providing insight into usage behavior. The study data and analysis presented here reflect an edited portion of the overall data from our hierarchical analysis process: It does not include information that would be reflective of any specific participant or patient's identity, nor information that was deemed irrelevant to study objectives by the coding team (i.e., pleasantries exchanged between interviewer and participant, or details of participants' workplaces or therapy techniques).

Participant Information: Clinician Background and Experience

Because purposive sampling criteria for inclusion in this study were broad, clinical experiences varied widely. Clinicians reported general clinical SLP experience ranging from 3 to 20 years. Clinician experience with UBT ranged from 1 to 7 years (M = 3.92 years, SD = 2.20 years), with clinicians reporting treating three to > 100 clients using UBT during that time. Five of the clinicians currently work in an academic research setting or university clinic and currently use UBT. One of the clinicians reported currently working in a high school, and another reported working in a skilled nursing facility; both of these clinicians do not currently use UBT. All clinicians practice in the United States across the Midwest, Northeast, and South. All clinicians reported using UBT primarily for the remediation of residual speech sound errors in children and young adults. Other less frequently reported uses included apraxia of speech, cleft palate, accent modification in adults learning English as a second language, and development of speech skills for adults learning a (non-English) second language. Clinicians had the most experience working with children over the age of 8 years using UBT; however, all clinicians had UBT experience with at least one adult client. Five of the clinicians interviewed had experience training or supervising students in the use of UBT.

Theme 1: Performance Expectancy

As noted above, *performance expectancy* was defined as the degree to which a person—in this case, a clinician—expected UBT to help them in their job. Across all interviews, there were 109 coded statements that were grouped into 18 categories (with three subcategories) in the "performance expectancy" theme. Table 2 displays the number of coded statements belonging to each category and subcategory ("occurrences (n)"), a description of the code, and the number of clinicians who contributed the statements ("participants (n)"). Note that one participant could produce multiple statements that were counted in a single category. For example, in Table 2, there were 15 occurrences of coded statements falling into the "benefit.visualization" category from seven participants. This means that seven clinicians contributed at least one statement that fell into "benefit.visualization" category, with 15 total occurrences of statements in that category across all participants (meaning that some people mentioned the topic more than once).

The top categories in the theme of "performance expectancy" were *benefit.visualization* and *perception*. UBT offers visualization of tongue movements in real time, which all clinicians reported as allowing them to

Table 2. Performance expectancy theme data.

Category	Subcategory	Description	Occurrences *n*	Participants *n*
Benefit.visualization		Being able to visualize the tongue was helpful for client, clinician, and family	15	7
Perception		Related to question about if clinicians rely on perception, image, or combination	13	7
Progress.motivation		Poor motivation might limit progress; good motivation might enhance progress	8	3
Benefit.treating without		Use of US facilitated how clinicians treat without US	7	6
Benefit.family interpretation		How (easily) families can understand the image	7	6
Benefit.progress		Related to quicker or better progress with US compared to traditional	7	5
Benefit.cues		US facilitates clinician cuing (with and without US)	7	4
Barrier.generalize		Related to lack of generalization following US intervention	7	3
Benefit.motor control		How US can illustrate progress at a motor control level	6	4
Barrier.engagement		Related to client boredom while using US	5	2
Benefit.professionalism		How US enhanced clinician professional development	4	4
Benefit.engagement	Benefit.explore	Related to how US allows for exploration of movement	4	4
	Benefit.curiosity	US facilitates client and parent interest	4	3
	Benefit.motivation	How US can enhance motivation	2	1
Benefit.repetition		Related to a specific question of how quickly clinicians can diagnose an articulatory pattern with US; clinicians identified that they can almost immediately see what's going on	3	3
Benefit.complexity		US allows clinicians to appreciate complexity of speech	3	2
Benefit.education		Related to how exposure to US on a student level (SLP students) can facilitate learning	2	2
Progress.evaluation		SLP uses clinical reasoning during evaluation to determine progress and potential for progress	2	2
Benefit.real-time		Real-time feedback facilitated progress	2	2
Benefit.dependency		Unique to one clinician; they feel they cannot provide treatment without US anymore	1	1

Note. US = ultrasound; SLP = speech-language pathologist.

appreciate the complexity and motor control aspects of speech while also improving their ability to translate that motor complexity to the client and their family:

> When your tongue's inside your mouth, and you're not looking at it, it would be very difficult to have someone understand how to move their tongue in a way that they have never done before. Whereas if you do it on an ultrasound and just show them, "I want you to move this here," they can do it. (P2)

Five clinicians stated that the visualization provided by ultrasound biofeedback facilitates job performance by allowing them to see and document incremental "hidden progress"—that is, progress at a motor control level that may not be apparent auditorily at a phonemic category level. As one clinician remarked:

> For the clinician and client, the feedback. Being able to know progress, like motor progress, even if perceptually it's still wrong, or inaccurate, you can keep track of that "hidden progress." (P6)

Separately, clinicians noted that UBT can facilitate positive aspects of the clinical encounter, such as client motivation, engagement, curiosity, and motor exploration.

All clinicians commented that ultrasound contributed positively to their professional development by enhancing their clinical skills and, for some, made them more marketable for their current positions. Additionally, the clinical skills in question were reported by clinicians to translate to other speech intervention approaches and improved their ability to provide meaningful cues to their clients, even beyond speech:

> I work with swallowing now. I've learned to give more explicit feedback to people when doing their exercises and things like that. (P2)

> I'm a voice researcher which means that people care about the larynx. People use ultrasound for the larynx, trying to understand how the larynx is shaped or moving to produce sound. (P5)

In the performance expectancy category, clinicians also noted some potential negative factors of ultrasound biofeedback. Three clinicians noted that client engagement can remain low when using ultrasound biofeedback based on the client's own experiences. That is, if clients have decreased motivation toward improving speech in general, clinicians felt those feelings carried forward to UBT. Furthermore, four clinicians noted that UBT suffers from the same challenges to generalization, or the extension of speech motor learning to situations outside a therapy context, as with any intervention approach: "That's something that ultrasound can't even solve" (P2).

Theme 2: Effort Expectancy

In Theme 2 ("effort expectancy"), clinicians identified a number of factors related to the ease of use of UBT. The highest number of categories (26, with 15 subcategories) and coded statements (197) was linked to this theme. The categories with the most occurrences of coded statements in the data were "qualifications," "experience. time," "experience.supervision," and "barrier.image quality" (see Table 3).

While some clinicians reported that UBT was easy to use, others expressed frustration with the blurriness of ultrasound images and subjectivity of image interpretation, noting that many structures not visualized by ultrasound can affect speech production (e.g., tongue tip, lips, or soft palate). Thus, clinicians frequently referred to having to "fill in missing information," which was described as difficult.

Individual client characteristics were also noted as increasing effort and limiting client progress when using UBT, noted in "progress.age," "progress.id," and "progress.dd" categories (see Appendix A), with examples of developmental disabilities, intellectual disabilities, young age, apraxia, anatomy (e.g., cleft palate), different language backgrounds (from American English), and attentional deficits:

> And how does that translate? So I don't know if my knowledge with a normal shaping vocal tract would translate to these children. (P5)

Clinicians are tasked with addressing additional client and family concerns with UBT (category: "concerns") that may be linked to its relative novelty as a speech intervention. For instance, two clinicians identified that they have had to address family concerns related to using a medical device, radiation,[1] heat, and discomfort. Clinicians also noted that they had to take additional infection control precautions with UBT due to having to carefully clean the ultrasound probe between clients. These concerns may also be affected by the COVID-19 pandemic (as was mentioned by many clinicians) and the increased awareness of infection possibilities in SLP clinical environments.

A variety of suggestions (category: "wants") were provided by clinicians related to improving the ease of use of UBT. Clinicians suggested equipping the ultrasound scanner with a drawing tool to mark the desired target tongue shape, adding animations to the image to improve client engagement, having multiple views of the tongue displayed concurrently, and combining UBT with another intervention or imaging approach. One clinician suggested combining the visual of UBT with tactile information:

> Something else I have always thought about is combining that visual aspect with some tactile because I rely on both tactile and visual a lot. Like a lot of times, the child looking at the screen does not have any idea how they need to raise their sides of the tongue, so I have to put that [ultrasound probe] aside and start using a tongue depressor trying to touch the side and then bring a model and show them, "Okay here are your sides. I want you to bring it up, try to touch your molars." (P1)

Clinicians also identified training demands associated with UBT as a factor in the overall effort required to adopt UBT. All clinicians interviewed felt that SLPs desiring to use UBT should have some additional training

[1]Clinicians noted that while they were aware that ultrasound does not expose patients to ionizing radiation, families were not always aware of this, and therefore, the clinicians needed to talk with families to assuage these concerns.

Table 3. Effort expectancy theme data.

Category	Subcategory	Description	Occurrences *n*	Participants *n*
Qualifications		Related to question of qualifications for clinicians using US	29	7
Experience.time		How long clinicians have used US	12	7
Barrier.training		Limited graduate student training, continuing education	11	6
Experience.supervision		Clinician experience with supervising SLP students	10	7
Barrier.image quality		Poor quality of image interfering with US utility	10	5
Variety		Specific quotes about how a variety of tongue shapes and individual differences can change treatment and visualization	9	6
Experience.n of clients		Number of US clients seen by clinicians interviewed	8	7
Experience.years total		Total years of practice or experience without/before US	7	7
Image.identification		Related to question of if clinicians can consistently identify what's going on in image	6	6
Progress.age		Age/maturity may impact progress	5	5
Progress.id		Intellectual disabilities may limit progress	5	4
Barrier.anatomy		Anatomy interfering with image quality (e.g., cleft palate)	5	3
Barrier.simpler		Related to usability and intuitiveness of display	5	2
Benefit.ease		Related to how US simplifies learning	5	2
Progress.dd		Developmental disabilities may limit progress	4	3
Barrier.subjectivity		Related to difficulty in determining clinical judgment of tongue position or accuracy of sound	4	1
Progress.attention		Attentional impairments (e.g., ADHD) may limit progress	3	3
Barrier.discomfort		Related to client discomfort when holding US probe or gel	3	2
Barrier.missing info		US is only able to capture the tongue; other factors might contribute to production	3	2
Progress.anatomy		Differences in anatomy can influence image and subsequent changes	3	2
Progress.motor control		Limitations in fine motor control may limit progress	3	1
Barrier.age		Limited ability to implement US due to client age	2	2
Concerns	Medical device	Related to anxiety using a medical device	3	3
	Radiation	Related to allaying concerns of (nonexistent) ionizing radiation from others	2	2
	Heat	Related to risk of heat	2	2
	Germs	Related to infection	1	1
	Discomfort	Related to physical discomfort	1	1
Wants	Animation	Desire for US to include an animated component or cartoon	3	3
	Vocal tract	Desire to increase visualization to entire vocal tract	3	2
	Draw	Desire for drawing tool	2	2
	Simpler	Desire for simpler display and controls	2	2
	Two views	Desire for multiple US views (e.g., coronal and sagittal)	2	2
	Age	Desire for US to relate to age of client	2	2
	Combination	Desire for US to be combined with other component (e.g., MRI, tactile)	2	2
	Portable	Desire for increased portability	1	1
	Engagement	Desire for US to be more engaging	1	1
	Image quality	Desire for improvement in image quality	1	1
Progress.apraxia		Children with apraxia may show limited progress	2	1

Note. US = ultrasound; SLP = speech-language pathologist; ADHD = attention-deficit/hyperactivity disorder; MRI = magnetic resonance imaging.

prior to implementing the intervention approach in their practice. Two clinicians felt that SLPs should treat a minimum number of clients with UBT under the supervision of an expert, while others felt that there should be a threshold of treatment contact hours to reach. As identified by one clinician, the quality of the instruction is important for determining eventual effectiveness in using UBT in practice:

> How effective, I should say. You can have a year of training, and if the training was bad, you will always be bad. (P6)

Theme 3: Social Influence

In the UTAUT model, the social influence theme refers to the degree to which a person believes that others, whom they deem to have high status, think they should use the technology. Our data included a number of comments by clinicians reflecting on broader social constructs, such as culture, diversity, and health equity hierarchies within and beyond treatment with UBT (50 coded statements across 11 categories). Following the suggestion of Bazeley (2009) that themes should incorporate elements constituting an explanatory model of participant experiences, comments relating to culture, diversity, and equity were assigned to the social influence theme, as these are inextricably linked to issues of esteem and status. The most common occurrences of coded statements were in the categories of "bias," "barrier.setting," and "barrier.cost for client" (see Table 4).

All clinicians discussed potential biases inherent to many rehabilitation interventions by virtue of many interventions being designed by and for majority groups. Non-White SLPs, compared to White clinicians, noted a lack of cultural responsivity present in the profession itself that is difficult to disentangle from UBT specifically:

> I will answer this coming from a multiple minority SLP. So overall, our field—our cultural responsiveness—to individuals who are linguistically diverse and have speech sound disorders, our field is very bad at it. Our clinicians have very little cultural responsiveness. So I don't see how that will not also bleed into an SLP who is treating a kid with a diverse linguistic background who might not [...] sound exactly how the White clinician would like for the kid to sound and then those biases and those preferences start to bleed into. (P6)

> Everyone produces certain sounds a little bit differently, especially, like we were talking about, the /r/ sound. So having that training [...] to be aware of those differences, not just how the speech sounds are produced, but cultural differences, dialectical differences, and what their norms are. Having a good idea of where they come from, what communities they're a part of, what cultures they're a part of. Having that foundation is definitely crucial, I think in not just ultrasound use, but in doing what we do. (P4).

Clinicians identified that the cost of UBT to the client and family can limit intervention opportunities for families of lower socioeconomic status. Intervention delivery in a university clinic may preclude the use of insurance (including Medicaid) and may be more expensive than alternative interventions, as noted by one clinician from their experience at their specific clinic:

Table 4. Social influence theme data.

Category	Description	Occurrences *n*	Participants *n*
Bias	Related to specific question about bias in "correctness" and suggestions to minimize bias	13	7
Barrier.setting	Limited settings in which ultrasound is currently used/implemented	11	7
Success.feedback	Rely on feedback from clinicians or parents	6	4
Success.understood	Related to intelligibility of speech	5	4
Barrier.network	A limited network of collaborators can make implementation challenging	5	2
Success.social	Related to social aspect of speech (e.g., less bullying, school participation)	4	4
Barrier.cost for client	Expense of ultrasound for client	2	2
Benefit.ownership	Related to facilitating autonomy	1	1
Concerns.culture	Related to cultural concerns using ultrasound	1	1
Success.meaningful words	Able to say important words (e.g., name, sibling name)	1	1
Progress.language diversity	Ultrasound may not be able to detect baseline for diverse speakers and adversely impact progress	1	1

> We charge more, $65 for a 45-minute session, whereas if it were traditional articulation, it would be $45 for a 45-minute session. And we don't bill insurance. (P3)

Relatedly, the same clinician noted that family ability to acquire transportation to the clinic can limit intervention delivery, and it should be noted that transportation can act as a barrier to any intervention, not just UBT:

> I see a lot of upper middle-class families that are able to afford and have the resources to get their kids to the therapy. (P3)

Clinicians noted the importance of social outcomes for clients related to the use of UBT. Specifically, clinicians noted how clients might feel more confident speaking in a school environment and/or increased confidence saying familiar words (e.g., sibling names). One clinician noted that UBT equips clients with autonomy over their speech:

> I remember feeling like the kids took a lot more ownership of what was going on because they had the power to make these images change on the ultrasound. Because, like I said, not all of them are motivated by the same thing. So you get some that are really insecure about the way that they speak and their different sound errors and then some that are just like, "Hey, it is what it is," and didn't care much, or it doesn't bother them as much. But the kind of power and ownership of knowing, "Okay, if I do this with my tongue, and I can see it, I can fix it." And I think that kind of fed into the success as well. (P4)

Speech was also discussed with socially influenced wording, such as terminology like "perfect" and "clear," which were connected with discussions of clients' personal factors such as accent, family, and age.

> I haven't gotten to work with a large diversity of accents with ultrasound. [...] I think taking into account the client's expectation and their functional communication is important.... And I guess what's in their zone of development, right. So like what's in their proximal zone of development. So if they come in and they have a super distorted /r/ and it's really impacting how well I understand them and they can get closer but not an exact /r/, I'm still happy with them making some progress because that might help them be better understood in a functional way. Often that's not the case though. Often you can pretty well understand them, they just have a noticeable /r/ error and the parents or the child want perfection, for lack of a better word. (P3)

Clinician discussion of professional networking was also categorized under the theme of social influence. Clinicians stated that professional networking was critical for their success with ultrasound. These networks could consist of other professionals using UBT and may include linguistically diverse individuals, researchers, or clinical supervisors—notably, people who would have traditionally collaborated in more limited ways. While our data cannot suggest that longstanding clinical research hierarchies are "flattened" in the face of a new technology such as UBT, these new professional networks empowered the clinicians in our study to creatively work together and learn from others' experiences and knowledge:

> Just not having a person who is experienced right on site, we're all just kind of learning together. (P7) But like having a network of other people, that's a big one, I would say. So recently I started talking more with [a clinical supervisor at a university]. And [...] we consult on clients and she has referred some of her clients who have been resistant to progress over to me just to be, like "we've been trying this for a while, maybe you try it," or "see what you think," as a consultation kind of thing. And that's really, I think, great too. To be able to talk to other people who are using the same technology as you. (P3)

> [Working with ultrasound] has also given me a lot of opportunities to work with linguistically diverse individuals. They tend to be relegated.... It has given me nice opportunities to connect to other SLPs or other researchers who might be using or might be interested in using ultrasound. (P6)

Theme 4: Facilitating Conditions

All clinicians discussed the structural and organizational support for UBT, or facilitating conditions, with 50 coded statements across six categories. Notably, clinicians reported that there are multiple infrastructure barriers to UBT as it currently stands. The most occurrences of coded statements were in categories "barrier.accessibility," "experience.settings," and "barrier.cost institution" (see Table 5).

All clinicians had experience using UBT in an academic university setting and had not seen it used elsewhere. For clinicians currently working outside academic and university settings, they reported the possible benefits that it could provide to a K–12 school setting, for example:

Table 5. Facilitating conditions theme data.

Category	Description	Occurrences *n*	Participants *n*
Barrier.accessibility	Limited availability/access to US and equipment (from client and clinician perspective)	15	7
Experience.settings	Setting in which clinicians used or saw US (might be current setting)	14	7
Barrier.cost institution	Expense of US for institution	9	4
Barrier.COVID	COVID-19 limited implementation of US	5	3
Experience.supervision.cont.use	Follow-up question related to whether supervised students have continued using US	4	3
Barrier.portable	Limited ability to transport	3	3

Note. US = ultrasound.

> But I think if you had one of those things in a school [. . .] the amount of children that would be off your caseload quickly just for the ease of being able to teach them a sound quickly. I mean—them learning the shape or the tongue positioning is like half the battle, you know. And it takes years. I mean, I definitely think that if schools invested in it, it would pay off. But, try explaining that to them. (P2)

With "try explaining that to them," the SLP communicates frustration at being unable to effectively advocate for a technology that represents an efficient solution to an SLP problem: tongue positioning in speech sound disorders. Lack of access to UBT in a school setting may be related to low funding in general for SLP services in this practice environment:

> We fight to get materials, we pay for our own books and our own supplies. We have what we have and that's kind of it. (P4)

Despite the number of clinicians interviewed who had experience supervising or teaching students, none of the clinicians were aware of former students continuing with the use of UBT upon becoming licensed practitioners, and this appears to be related largely to lack of institutional support:

> Students usually end up going to private practice or schools or hospitals, which as we talked about earlier, they don't have that type of instrumentation in those places [. . .] I wish they could use it, but I don't think so. I'm pretty sure 99.95% that they are not using later on. (P1)

Accessibility was the top-mentioned barrier to implementation of UBT, with every clinician mentioning it at least once in their interview. Clinicians noted lack of access as a barrier from multiple perspectives, including institutional and client. Clinicians also noted that many of their SLP peers may not even be aware of the existence of the technology. Clinicians were aware that the current pattern of UBT restriction to the clinical environment, that is, academic and hospital environments, is connected to the historically low portability of ultrasound devices and mentioned that this reduces its effectiveness for home practice. It was noted that transition to telehealth in response to the COVID-19 pandemic presented a unique challenge for many clinicians currently using UBT and largely eliminated UBT from their intervention toolboxes at least temporarily.

Discussion

In this qualitative study, we aimed to explore clinician experiences with UBT for speech sound disorders. We used the UTAUT model to thematically organize clinician responses related to usage behavior of the technology. Most coded responses were sorted into the "effort expectancy" theme, followed by the "performance expectancy," "social influence," and "facilitating conditions" themes. In the following sections, we discuss the four thematic areas of the UTAUT model presented in order of code frequencies and interpret how these results inform our understanding of SLP decision making and usage behavior toward a new technology such as UBT.

Prioritizing Effort Expectancy

Overall, from the number and detail of their responses under the theme of effort expectancy, clinicians appeared to prioritize "ease of use" considerations when making clinical decisions regarding UBT. This corresponds with clinical priorities; a systematic review of technology-based interventions in health care professional

practice found "ease of use" to be a major factor in the uptake of technology in clinical settings (Keyworth et al., 2018). SLPs, like other health care professionals, face numerous pressures in clinical settings, such as increasing caseload sizes, productivity demands, and paperwork, as well as generally underresourced operating conditions (Kalkhoff & Collins, 2012). Considerations related to "ease of use" reflect these mounting pressures: SLPs in this study seemed to demonstrate an awareness of these demands as they commonly reflected on the need for SLPs to receive specialized, high-quality training prior to using ultrasound clinically. Furthermore, the clinicians interviewed who practice outside the academic university setting acknowledged how the barriers of a particular setting (e.g., caseload size and limited resources) can complicate more widespread uptake of ultrasound technology.

This apparent focus on "ease of use" may also reflect the fact that our sample represented the overwhelmingly female-gendered nature of the speech-language pathology profession (ASHA, 2020). The UTAUT construct of "effort expectancy" has been suggested as more salient for women than for men due to socially structured gender roles: Men often prioritize information about productivity when making decisions involving technology, but women tend to incorporate many sources of information in their decisions, including information about the effort a technology will take to use (Venkatesh & Morris, 2000).

Along with "gender," "age" and "experience" are suggested to moderate the relationship between "effort expectancy" and behavioral intention to use a technology, with older or less experienced individuals being more likely to believe that a technology will require effort to use (Venkatesh et al., 2003). For our sample of clinicians, experience (as they themselves defined it by number of patients, years of clinical work, and amount of training) were all noted important factors for successful ultrasound use. Though a relatively novel technology when compared to more common systems such as modified barium swallow studies, ultrasound appears to require similar experiences to promote a sense of clinical self-efficacy: a combination of both formal educational preparation and real clinical experience (Caesar & Kitila, 2020). Prior clinical experience is a cornerstone of SLP decision-making processes for children with speech sound disorders (Furlong et al., 2018). Although current SLP students were not interviewed for this study, we might expect that students would be more likely to perceive ultrasound to be more effortful than more experienced SLPs. While it is not possible to investigate age effects in this qualitative study, as all participants were near the same age (30s and early 40s), we suspect that age would moderate effort expectancy and behavioral intention in similar ways in the speech-language pathology group as in other groups using the UTAUT model.

Performance Expectancy: Meeting Clients' Needs

"Performance expectancy," or the belief that ultrasound would improve job performance, was the second most discussed theme among SLPs in this study, which corresponds to predictions from the UTAUT model, in which "effort expectancy" and "performance expectancy" explain most of the variance in quantitative data on user acceptance of technology (Venkatesh et al., 2003). While the "effort expectancy" theme was generally reflective of clinicians thinking about themselves in the context of their practice, "performance expectancy" often related to the client's needs or preferences, ultimately helping the clinician to better perform their job. Performance, for the SLPs in this study, was recognized when a client was able to accomplish a personally meaningful outcome. Many clinicians sought to contextualize the intervention with salient outcomes (e.g., client saying sibling names) and valued functional communication and client/family preferences. Some clinicians described UBT as motivating for clients, resulting in increased client "buy-in" or improved disposition to participate in the therapy program. One clinician identified that UBT equipped children with feelings of autonomy, and the clinician associated those feelings of empowerment with improved outcomes. In research that aims to quantify technology acceptance on the part of clinicians, models such as UTAUT may be complemented by measures of a client's feeling of empowerment (Brew-Sam, 2020), since both the clinician and client are implicated in effective use of a technology. In the Bacsfalvi (2007) qualitative study of ultrasound used for habilitation of hard of hearing speakers, clients noted a sense of enjoyment and motivation associated with improved control of perception when using ultrasound: "[Ultrasound] tells you what you are missing from your speech" and "This way, I can be successful. I can pass *(he means be understood by unfamiliar hearing speakers)*" (p. 215). Future research may focus on defining aspects of this promising link between a sense of control and motivation to participate in UBT.

Many of the comments on performance expectancy reflect the challenge of integrating a new technology into the established medical model of service provision, which focuses on diagnosis and evidence-based treatment methods from a narrowly defined participant group. Clinicians readily discussed UBT in terms of diagnosis and medical signs and symptoms, with speculation about its suitability for a range of clinical populations, but expressed some uncertainty about expanding use of UBT for individuals who did not meet particular criteria from treatment studies. Clinicians were particularly concerned with the lack of specific research examining the efficacy of UBT for specific speech

disorders, languages, and dialects and perceived this as a definite limitation on its use for those populations. At the same time, the clinicians in this study mentioned other valuable aspects of UBT (e.g., enhanced interactions between therapist and client) that entered into their clinical decision making regarding its adoption. Clinicians could imagine opportunities offered by ultrasound for improved treatment for various motor-based disorders while at the same time highlighting perceived barriers and challenges to implementing these technologies in established practice patterns.

A Call for Greater Racial and Ethnic Representation in UBT Research

Within the "social influence" theme, clinicians (particularly those from underrepresented groups) mentioned the importance of creating culturally sensitive interventions and improving SLP cultural responsiveness, even extending beyond the bounds of UBT. These words echo the calls from many speech-language researchers for greater representation of racially and ethnically diverse individuals in education and postgraduate training experiences (Maldonado et al., 2019; Narayanan & Ramsdell, 2022; Quach & Tsai, 2017). There has been little research in SLP focusing on the diversity of speech sound and articulatory phenomena in individuals who are from underrepresented groups and even less on the issues highlighted by ultrasound imaging. Currently, most ultrasound research on speech diversity occurs within the field of linguistics (e.g., Lawson et al., 2013; a sociophonetic ultrasound study of tongue movement in Scottish dialects). Breaking down the artificial disciplinary divide between linguistics and speech-language pathology could assist in achieving goals of representative research for all social and cultural groups.

Additionally, in assessment and treatment, when there is an SLP–client mismatch in linguistic and/or cultural background, best practices call for involving the family of the client, where possible. Family members can provide information about their community's linguistic norms, as well as give information about language experiences and preferences, values, family roles and responsibilities, and environmental context (McLeod et al., 2017). Some SLPs interviewed in this study referred to the practice of incorporating parents in the ultrasound therapy session, noting beneficial outcomes when parents were able to participate in the ultrasound therapy context and understand the experiences of their children.

Structural Organizations Act as Facilitating Conditions

Structural support for technology, or "facilitating conditions," contributed to usage behavior of UBT. When beginning to work with ultrasound technology for speech therapy, the SLPs in this study sought information from formal training. For those who continued to work with the technology, conversations with colleagues who were using ultrasound assisted SLPs in clinical decision making. These findings are consistent with results from the study of Zipoli et al. (2005), in which clinician experience and colleague opinions are prioritized in clinical decision making over research studies. These kinds of conversations may improve the clinical translation of a technology that has been restricted previously to research settings and help clinicians identify opportunities for innovative implementation of UBT techniques. The limited network currently available for UBT reported by many interviewees may partly explain the reduced uptake of the technology in other settings. A structural organization (e.g., setting, network) that is supportive of the adoption of innovative health care delivery, including novel technology, may be critical for the implementation and facilitation of shared clinical decision making and active client involvement in care (Fisher et al., 2016).

Limitations

The results of this study should be interpreted with knowledge of its limitations. While our sample was diverse across multiple personal factors, we do acknowledge that the views expressed in this study may not be fully representative of all clinicians and researchers using UBT, including those outside the United States. Because our clinician participants were based in the United States, comments specific to billing and health care (i.e., Medicaid) may not apply to other clinician experiences in other countries. The presence of an interviewer creates an inherent limitation in all qualitative interview research, as there is a risk of social desirability bias, and clinicians may have felt compelled to provide responses compatible with socially acceptable norms. Positively, the interviewer in this study was unknown to the participants and was from a different allied health field, which we believe facilitated increased transparency of responses from the interviewees. Additionally, factors of gender, age, and experience are known to moderate the effect of "effort expectancy" in the UTAUT model; however, because our sample was fairly homogenous in age and experience, we are not able to fully explore the impact of these factors on the use of UBT.

Conclusions

In their interviews on the use of UBT for speech treatment, SLPs gave relative prominence to concepts that related to "effort expectancy" and "performance expectancy"—results that correspond with the UTAUT

model, where these two concepts previously explained most of the variance in a quantitative data set (Venkatesh et al., 2003). Clinicians spoke favorably about using ultrasound for speech sound disorder treatment but acknowledged institutional barriers, as well as reflecting on the need for more UBT research including participants in underrepresented groups. SLPs who were from underrepresented groups themselves connected these personal factors to working with UBT as clinicians and researchers. Clinicians also reflected on the valuable role that high-quality training and collaborations with colleagues played in their decision-making processes regarding UBT.

In addition to the constructs "effort expectancy" and "performance expectancy," personal factors such as gender and race/ethnicity and environmental factors such as institutional setting and collegial environment emerged as being critical to SLPs' decision making about UBT. Their importance suggests that an expanded definition of the UTAUT construct "social influence," as portrayed in this study, may assist in understanding factors influencing technology use for SLPs. Additionally, such factors may be important moderators of relationships among constructs comprising the UTAUT model in future mixed-methods studies of SLP technology acceptance.

Data Availability Statement

The available original contributions presented in the study are included in the article. Further inquiries can be directed to the corresponding author.

Acknowledgments

Research presented in this article was supported in part by a grant from the National Institutes of Health (R01 DC017301). The funder played no role in the design or conduct of this study nor the writing of this article. The authors thank the speech-language pathologists who so generously contributed their time and talents to this project in the form of interviews and survey responses.

References

Adler-Bock, M., Bernhardt, B. M., Gick, B., & Bacsfalvi, P. (2007). The use of ultrasound in remediation of North American English /r/ in 2 adolescents. *American Journal of Speech-Language Pathology, 16*(2), 128–139. https://doi.org/10.1044/1058-0360(2007/017)

Ajzen, I., & Fishbein, M. (1980). *Understanding attitudes and predicting social behavior*. Prentice Hall.

American Speech-Language-Hearing Association. (2020). *2020 Member and affiliate profile: Annual demographic and employment data*. https://www.asha.org/siteassets/surveys/2020-member-and-affiliate-profile.pdf

Bacsfalvi, P. (2007). *Visual feedback technology with a focus on ultrasound: The effects of speech habilitation for adolescents with sensorineural hearing loss* [Doctoral dissertation, University of British Columbia]. University of British Columbia Library. https://open.library.ubc.ca/collections/ubctheses/831/items/1.0100430

Bacsfalvi, P. (2010). Attaining the lingual components of /r/ with ultrasound for three adolescents with cochlear implants. *Canadian Journal of Speech-Language Pathology and Audiology, 34*(3), 206–218.

Baribeau, Y., Sharkey, A., Chaudhary, O., Krumm, S., Fatima, H., Mahmood, F., & Matyal, R. (2020). Handheld point-of-care ultrasound probes: The new generation of POCUS. *Journal of Cardiothoracic and Vascular Anesthesia, 34*(11), 3139–3145. https://doi.org/10.1053/j.jvca.2020.07.004

Basit, T. (2003). Manual or electronic? The role of coding in qualitative data analysis. *Educational Research, 45*(2), 143–154. https://doi.org/10.1080/0013188032000133548

Bazeley, P. (2009). Analysing qualitative data: More than 'identifying themes.' *Malaysian Journal of Qualitative Research, 2*(2), 6–22.

Bernhardt, B., Gick, B., Bacsfalvi, P., & Ashdown, J. (2003). Speech habilitation of hard of hearing adolescents using electropalatography and ultrasound as evaluated by trained listeners. *Clinical Linguistics & Phonetics, 17*(3), 199–216. https://doi.org/10.1080/0269920031000071451

Bernhardt, M. B., Bacsfalvi, P., Adler-Bock, M., Shimizu, R., Cheney, A., Giesbrecht, N., O'Connell, M., Sirianni, J., & Radanov, B. (2008). Ultrasound as visual feedback in speech habilitation: Exploring consultative use in rural British Columbia, Canada. *Clinical Linguistics & Phonetics, 22*(2), 149–162. https://doi.org/10.1080/02699200701801225

Boyce, S. (2015). The articulatory phonetics of /r/ for residual speech errors. *Seminars in Speech and Language, 36*(4), 257–270. https://doi.org/10.1055/s-0035-1562909

Boyce, S., & Lee, S. (2016, September). *Ultrasound for speech-language pathology* [Presentation]. World Congress of Ultrasound, Texas Tech University, Lubbock, TX.

Brackenbury, T., Burroughs, E., & Hewitt, L. E. (2008). A qualitative examination of current guidelines for evidence-based practice in child language intervention. *Language, Speech, and Hearing Services in Schools, 39*(1), 78–88. https://doi.org/10.1044/0161-1461(2008/008)

Brew-Sam, N. (2020). Discussion and deriving research gaps. In N. Brew-Sam (Ed.), *App use and patient empowerment in diabetes self-management: Advancing theory-guided mHealth research* (pp. 243–264). Springer Fachmedien Wiesbaden. https://doi.org/10.1007/978-3-658-29357-4_15

Caesar, L. G., & Kitila, M. (2020). Speech-language pathologists' perceptions of their preparation and confidence for providing dysphagia services. *Perspectives of the ASHA Special Interest Groups, 5*(6), 1666–1682. https://doi.org/10.1044/2020_PERSP-20-00115

Chan, F. K., Thong, J. Y., Venkatesh, V., Brown, S. A., Hu, P. J., & Tam, K. Y. (2010). Modeling citizen satisfaction with mandatory adoption of an e-government technology. *Journal of the Association for Information Systems, 11*(10), 519–549. https://doi.org/10.17705/1jais.00239

Cheng, Y. S., Yu, T. F., Huang, C. F., Yu, C., & Yu, C. C. (2011). The comparison of three major occupations for user acceptance of information technology: Applying the UTAUT model. *iBusiness, 03*(2), 147–158. https://doi.org/10.4236/ib.2011.32021

Cleland, J., Scobbie, J. M., Roxburgh, Z., Heyde, C., & Wrench, A. (2019). Enabling new articulatory gestures in children with persistent speech sound disorders using ultrasound visual biofeedback. *Journal of Speech, Language, and Hearing Research, 62*(2), 229–246. https://doi.org/10.1044/2018_JSLHR-S-17-0360

Cleland, J., Scobbie, J. M., & Wrench, A. A. (2015). Using ultrasound visual biofeedback to treat persistent primary speech sound disorders. *Clinical Linguistics & Phonetics, 29*(8–10), 575–597. https://doi.org/10.3109/02699206.2015.1016188

Clevert, D.-A., Schwarze, V., Nyhsen, C., D'Onofrio, M., Sidhu, P., Brady, A. P., & European Society of Radiology. (2019). ESR statement on portable ultrasound devices. *Insights Into Imaging, 10*(1), 89. https://doi.org/10.1186/s13244-019-0775-x

Diepeveen, S., van, H. L., Terband, H., denn, S. B., & Maassen, B. (2020). Clinical reasoning for speech sound disorders: Diagnosis and intervention in speech-language pathologists' daily practice. *American Journal of Speech-Language Pathology, 29*(3), 1529–1549. https://doi.org/10.1044/2020_AJSLP-19-00040

Dugan, S., Li, S., Masterson, J., Woeste, H., Mahalingam, N., Spencer, C., Mast, T. D., Riley, M., & Boyce, S. (2019). Tongue part movement trajectories for /r/ using ultrasound. *Perspectives of the ASHA Special Interest Groups, 4*(6), 1644–1652. https://doi.org/10.1044/2019_PERS-19-00064

Falkowski, A. L., Jacobson, J. A., Freehill, M. T., & Kalia, V. (2020). Hand-held portable versus conventional cart-based ultrasound in musculoskeletal imaging. *Orthopaedic Journal of Sports Medicine, 8*(2), 2325967119901017. https://doi.org/10.1177/2325967119901017

Fishbein, M., & Ajzen, I. (1975). *Belief, attitude, intention, and behavior*. Addison-Wesley.

Fisher, E. S., Shortell, S. M., & Savitz, L. A. (2016). Implementation science: A potential catalyst for delivery system reform. *JAMA, 315*(4), 339–340. https://doi.org/10.1001/jama.2015.17949

Furlong, L., Serry, T., Erickson, S., & Morris, M. E. (2018). Processes and challenges in clinical decision-making for children with speech-sound disorders. *International Journal of Language & Communication Disorders, 53*(6), 1124–1138. https://doi.org/10.1111/1460-6984.12426

Gick, B., Bacsfalvi, P., Bernhardt, B. M., Oh, S., Stolar, S., & Wilson, I. (2008). A motor differentiation model for liquid substitutions in children's speech. *Proceedings of Meetings on Acoustics, 1,* 10.

Hardcastle, W. J., & Gibbon, F. (1997). Electropalatography and its clinical applications. *Instrumental Clinical Phonetics,* 149–193. https://doi.org/10.1002/9780470699119.ch6

Hennington, A., & Janz, B. D. (2007). Information systems and healthcare XVI: Physician adoption of electronic medical records: Applying the UTAUT model in a healthcare context. *Communications of the Association for Information Systems, 19*. https://doi.org/10.17705/1CAIS.01905

Jeng, D. J.-F., & Tzeng, G.-H. (2012). Social influence on the use of clinical decision support systems: Revisiting the unified theory of acceptance and use of technology by the fuzzy DEMATEL technique. *Computers & Industrial Engineering, 62*(3), 819–828. https://doi.org/10.1016/j.cie.2011.12.016

Kalkhoff, N. L., & Collins, D. R. (2012). Speech-language pathologist job satisfaction in school versus medical settings. *Language, Speech, and Hearing Services in Schools, 43*(2), 164–175. https://doi.org/10.1044/0161-1461(2011/11-0007)

Katz, W. F., & Mehta, S. (2015). Visual feedback of tongue movement for novel speech sound learning. *Frontiers in Human Neuroscience, 9*. https://doi.org/10.3389/fnhum.2015.00612

Keyworth, C., Hart, J., Armitage, C. J., & Tully, M. P. (2018). What maximizes the effectiveness and implementation of technology-based interventions to support healthcare professional practice? A systematic literature review. *BMC Medical Informatics and Decision Making, 18*(1), 93–21. https://doi.org/10.1186/s12911-018-0661-3

Kim, S., Lee, K.-H., Hwang, H., & Yoo, S. (2016). Analysis of the factors influencing healthcare professionals' adoption of mobile electronic medical record (EMR) using the unified theory of acceptance and use of technology (UTAUT) in a tertiary hospital. *BMC Medical Informatics and Decision Making, 16*(1), 12. https://doi.org/10.1186/s12911-016-0249-8

Kohnke, A., Cole, M. L., & Bush, R. (2014). Incorporating UTAUT predictors for understanding home care patients' and clinician's acceptance of healthcare telemedicine equipment. *Journal of Technology Management & Innovation, 9*(2), 29–41. https://doi.org/10.4067/S0718-27242014000200003

Kuschmann, A., Nayar, R. V., Lowit, A., & Dunlop, M. (2021). The use of technology in the management of children with phonological delay and adults with acquired dysarthria: A U.K. survey of current speech-language pathology practice. *International Journal of Speech-Language Pathology, 23*(2), 145–154. https://doi.org/10.1080/17549507.2020.1750700

Lawson, E., Scobbie, J. M., & Stuart-Smith, J. (2013). Bunched /r/ promotes vowel merger to schwar: An ultrasound tongue imaging study of Scottish sociophonetic variation. *Journal of Phonetics, 41*(3–4), 198–210. https://doi.org/10.1016/j.wocn.2013.01.004

Liu, J., Xu, J., Forsberg, F., & Liu, J-B. (2019). *CMUT/CMOS-based Butterfly iQ—A portable personal sonoscope. Department of Radiology Faculty Papers, Paper 72*. https://jdc.jefferson.edu/radiologyfp/72

Maldonado, C., Ashe, A., Bubar, K., & Chapman, J. (2019). American speech-language pathologists' training and legislative knowledge when working with immigrant and refugee populations: A qualitative study. *Perspectives of the ASHA Special Interest Groups, 4*(5), 1148–1161. https://doi.org/10.1044/2019_PERS-SIG17-2019-0028

McAllister, L., & Rose, M. (2000). Speech-language pathology students: Learning clinical reasoning. In J. Higgs & M. Jones (Eds.), *Clinical reasoning in the health professions* (pp. 205–213). Butterworth-Heinemann.

McAllister Byun, T., & Campbell, H. (2016). Differential effects of visual-acoustic biofeedback intervention for residual speech errors. *Frontiers in Human Neuroscience, 10,* 567. https://doi.org/10.3389/fnhum.2016.00567

McAllister Byun, T., Hitchcock, E. R., & Swartz, M. T. (2014). Retroflex versus bunched in treatment for rhotic misarticulation: Evidence from ultrasound biofeedback intervention. *Journal of Speech, Language, and Hearing Research, 57*(6), 2116–2130. https://doi.org/10.1044/2014_JSLHR-S-14-0034

McLeod, S., Verdon, S., Baker, E., Ball, M. J., Ballard, E., David, A. B., Bernhardt, B. M., Bérubé, D., Blumenthal, M., Bowen, C., Brosseau-Lapré, F., Bunta, F., Crowe, K., Cruz-Ferreira, M., Davis, B., Fox-Boyer, A., Gildersleeve-Neumann, C., Grech, H., Goldstein, B., . . . Zharkova, N. (2017). Tutorial: Speech assessment for multilingual children who do not speak the same language(s) as the speech-language pathologist. *American Journal of Speech-Language Pathology, 26*(3), 691–708. https://doi.org/10.1044/2017_AJSLP-15-0161

Modha, G., Bernhardt, B. M., Church, R., & Bacsfalvi, P. (2008). Case study using ultrasound to treat /ɹ/. *International Journal of Language & Communication Disorders, 43*(3), 323–329. https://doi.org/10.1080/13682820701449943

Narayanan, T. L., & Ramsdell, H. L. (2022). Self-reported confidence and knowledge-based differences between multilingual

and monolingual speech-language pathologists when serving culturally and linguistically diverse populations. *Perspectives of the ASHA Special Interest Groups 7*(1), 209–228. https://doi.org/10.1044/2021_PERSP-21-00169

Newell, W. H. (1994). Designing interdisciplinary courses. *New Directions for Teaching and Learning, 1994*(58), 35–51. https://doi.org/10.1002/tl.37219945804

Nielsen, J., & Landauer, T. K. (1993). A mathematical model of the finding of usability problems. *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems—CHI '93* (pp. 206–213). https://doi.org/10.1145/169059.169166

Palinkas, L. A., Horwitz, S. M., Green, C. A., Wisdom, J. P., Duan, N., & Hoagwood, K. (2015). Purposeful sampling for qualitative data collection and analysis in mixed method implementation research. *Administration and Policy in Mental Health, 42*(5), 533–544. https://doi.org/10.1007/s10488-013-0528-y

Popay, J., Rogers, A., & Williams, G. (1998). Rationale and standards for the systematic review of qualitative literature in health services research. *Qualitative Health Research, 8*(3), 341–351. https://doi.org/10.1177/104973239800800305

Preston, J. L., Brick, N., & Landi, N. (2013). Ultrasound biofeedback treatment for persisting childhood apraxia of speech. *American Journal of Speech-Language Pathology, 22*(4), 627–643. https://doi.org/10.1044/1058-0360(2013/12-0139)

Preston, J. L., Holliman-Lopez, G., & Leece, M. C. (2018). Do participants report any undesired effects in ultrasound speech therapy? *American Journal of Speech-Language Pathology, 27*(2), 813–818. https://doi.org/10.1044/2017_AJSLP-17-0121

Preston, J. L., Leece, M. C., & Maas, E. (2017). Motor-based treatment with and without ultrasound feedback for residual speech sound errors. *International Journal of Language & Communication Disorders, 52*(1), 80–94. https://doi.org/10.1111/1460-6984.12259

Preston, J. L., Leece M. C., McNamara, K., & Maas, E. (2017). Variable practice to enhance speech learning in ultrasound biofeedback treatment for childhood apraxia of speech: A single case experimental study. *American Journal of Speech-Language Pathology, 26*(3), 840–852. https://doi.org/10.1044/2017_AJSLP-16-0155

Preston, J. L., McAllister, T., Phillips, E., Boyce, S., Tiede, M., Kim, J. S., & Whalen, D. H. (2018). Treatment for residual rhotic errors with high- and low-frequency ultrasound visual feedback: A single-case experimental design. *Journal of Speech, Language, and Hearing Research, 61*(8), 1875–1892. https://doi.org/10.1044/2018_JSLHR-S-17-0441

Preston, J., McAllister Byun, T., Boyce, S., Hamilton, S., Tiede, M., Phillips, E., Rivera-Campos, A., & Whalen, D. (2017). Ultrasound images of the tongue: A tutorial for assessment and remediation of speech sound errors. *Journal of Visualized Experiments,* (119), Article e55123. https://doi.org/10.3791/55123

Quach, W., & Tsai, P.-T. (2017). Preparing future SLPs for the clinical world of cultural-linguistic diversity. *Perspectives of the ASHA Special Interest Groups, 2*(14), 82–102. https://doi.org/10.1044/persp2.SIG14.82

Rampin, R., & Rampin, V. (2021). Taguette: Open-source qualitative data analysis. *The Journal of Open Source Software, 6*(68), 3522. https://doi.org/10.21105/joss.03522

Roxburgh, Z., Scobbie, J. M., & Cleland, J. (2015). Articulation therapy for children with cleft palate using visual articulatory models and ultrasound biofeedback. *Proceedings of the 18th International Congress of Phonetic Sciences*. University of Glasgow. https://www.internationalphoneticassociation.org/icphs-proceedings/ICPhS2015/Papers/ICPHS0858.pdf

Saunders, B., Sim, J., Kingstone, T., Baker, S., Waterfield, J., Bartlam, B., Burroughs H., & Jinks, C. (2018). Saturation in qualitative research: Exploring its conceptualization and operationalization. *Quality & Quantity, 52*(4), 1893–1907. https://doi.org/10.1007/s11135-017-0574-8

Schwartz, A., & Elstein, A. S. (2008). Clinical reasoning in medicine. In J. Higgs & M. Jones (Eds.), *Clinical reasoning in the health professions* (pp. 223–234). Butterworth-Heinemann.

Smith, J. A. (2004). Reflecting on the development of interpretative phenomenological analysis and its contribution to qualitative research in psychology. *Qualitative Research in Psychology, 1*(1), 39–54. https://doi.org/10.1191/1478088704qp004oa

Sugden, E., Lloyd, S., Lam, J., & Cleland, J. (2019). Systematic review of ultrasound visual biofeedback in intervention for speech sound disorders. *International Journal of Language & Communication Disorders, 54*(5), 705–728. https://doi.org/10.1111/1460-6984.12478

Tong, A., Sainsbury, P., & Craig, J. (2007). Consolidated Criteria for Reporting Qualitative Research (COREQ): A 32-item checklist for interviews and focus groups. *International Journal for Quality in Health Care, 19*(6), 349–357. https://doi.org/10.1093/intqhc/mzm042

VanderKaay, S., Moll, S. E., Gewurtz, R. E., Jindal, P., Loyola-Sanchez, A., Packham, T. L., & Lim, C. Y. (2018). Qualitative research in rehabilitation science: Opportunities, challenges, and future directions. *Disability and Rehabilitation, 40*(6), 705–713. https://doi.org/10.1080/09638288.2016.1261414

Venkatesh, V., & Morris, M. G. (2000). Why don't men ever stop to ask for directions? Gender, social influence, and their role in technology acceptance and usage behavior. *MIS Quarterly, 24*(1), 115–139. https://doi.org/10.2307/3250981

Venkatesh, V., Morris, M. G., Davis, G. B., & Davis, F. D. (2003). User acceptance of information technology: Toward a unified view. *MIS Quarterly, 27*(3), 425–478. https://doi.org/10.2307/30036540

Wang, X., Liu, Y., & Liu, H. (2020). Examining users' adoption of precision medicine: The moderating role of medical technical knowledge. *International Journal of Environmental Research and Public Health, 17*(3), 1113. https://doi.org/10.3390/ijerph17031113

West, R. E. (2016) Breaking down walls to creativity through interdisciplinary design. *Educational Technology, 56,* 47–52.

Wills, M. J., El-Gayar, O. F., & Bennett, D. (2008). Examining healthcare professionals' acceptance of electronic medical records using UTAUT. *Issues in Information Systems, 9*(2), 396–401.

Yang, K. (2010). Determinants of US consumer mobile shopping services adoption: Implications for designing mobile shopping services. *Journal of Consumer Marketing, 27*(3), 262–270. https://doi.org/10.1108/07363761011038338

Zhou, T., Lu, Y., & Wang, B. (2010). Integrating TTF and UTAUT to explain mobile banking user adoption. *Computers in Human Behavior, 26*(4), 760–767. https://doi.org/10.1016/j.chb.2010.01.013

Zipoli, R. P., Jr., & Kennedy, M. (2005). Evidence-based practice among speech-language pathologists: Attitudes, utilization, and barriers. *American Journal of Speech-Language Pathology, 14*(3), 208–220. https://doi.org/10.1044/1058-0360(2005/021)

Appendix A (p. 1 of 2)

Interview Question List Shared With Clinician Participants Prior to Interview

Interview Question List for Clinicians:

PURPOSE: The purpose of this interview is to understand your past experiences with ultrasound for speech sound therapy. Questions will address your prior use of ultrasound, perceived barriers to usage, and benefits. The interview will last 1 hour.

PART 1: General Background

1. Open-ended: Tell me a little bit about your experiences using ultrasound.
   a.) Follow-up questions:
      i. How long have you used ultrasound?
      ii. How many children (approximately) have you seen using ultrasound?
      iii. How many years of experience treating speech sound disorders did you have before using ultrasound?
      iv. How many different speech disorders/diagnoses have you treated using ultrasound? Can you name some of them? What are the common ages for which you use ultrasound? Is there a particular setting where ultrasound is used frequently?
      v. Have you ever trained or supervised students who were using ultrasound?
      vi. Is there a "learning curve" in using ultrasound? What do you think are the minimum requirements for a clinician to have in order to use ultrasound? Do you believe that all clinicians using ultrasound meet these qualifications?

PART 2: "Pain Points"

2. Open-ended: What are the barriers to using ultrasound—for you, for students, and for clients?
   a.) Follow-up questions:
      i. Do you feel like you can consistently identify what is going wrong in articulation from the ultrasound?
      ii. How do you determine the "correctness" of a sound or an articulation? Do you feel like you do this reliably?
      iii. Do you think that the "correctness" of a sound is biased toward certain dialects or accents?
      iv. Suggestions to limit the bias in determining sound "correctness"? Does ultrasound objectify this decision making or is it also biased?
      v. Let's talk about those "in between" productions. Do you ever find yourself closing your eyes or looking away from the screen to try to understand if a production was correct or incorrect? If so, why?
      vi. Is it easy to describe what is going on in the image to parents? To student clinicians? To children? How easy is it to report progress?
      vii. What percentage of feedback given to children is KP vs. KR?
         (1) KP = knowledge of performance (characteristics of movement)
         (2) KR = knowledge of results (end outcome)
      viii. Is it easy for you to synthesize everything that was happening across ultrasound sessions for reporting purposes?
      ix. Reflecting on your clinical judgment and experiences, are there certain qualitative things (behaviors, movements, other diagnoses) that you would see in clients that would suggest to you that they will make slower (or no) progress?
      x. In your experience, what percentage of practice trials have to be repeated so that you can interpret what is happening with the tongue?
      xi. Do you find the ultrasound system flexible? Is there anything you would like to be able to control about the display? Any information that you would like to obtain to share with families but currently are not able to?
      xii. In your experience, do children find it easy to change their movements based on the ultrasound display?
      xiii. Do you think that sometimes there is progress going on at a motor control level that isn't noticeable yet in the sound of the production?
      xiv. Does ultrasound allow children to explore movement solutions that may facilitate sound production?

Appendix A (p. 2 of 2)

Interview Question List Shared With Clinician Participants Prior to Interview

xv. How does productivity or billing/reimbursement play a role in your use of ultrasound?

xvi. Do time constraints (e.g., set-up) ever limit your use of ultrasound? Do you ever make intervention decisions based on time?

xvii. How do you document a rationale for using ultrasound? How does documentation time vary when using ultrasound compared to other interventions?

xviii. If ultrasound was applied to a gamified format, what percent success rate would you consider ideal to maintain participant motivation while allowing room for improvement and feedback?

PART 3: Benefits

3. Open-ended: What, in your view, are the major benefits that ultrasound offers a clinician?

a) Follow-up questions:

i. How did you use the information from ultrasound to change the way you treated speech sound disorders (SSDs)?

ii. Did ultrasound change the way you approached SSDs (when treating w/out ultrasound)?

iii. What were some of the major things you learned about disorders through your experience with ultrasound?

iv. What were some of the major things you learned about clients through your experience with ultrasound?

v. How is client progress different when using ultrasound compared to other therapy methods?

vi. How does use of ultrasound contribute to your professional development and your job performance?

PART 4: Person- and Family-Centered Experience

4. Open-ended: What kinds of things do children and parents say about the ultrasound?

a) Follow-up questions:

i. Do children and their families “buy-in” to the concept of ultrasound? How do you overcome any hesitancy?

ii. How do you explain the concept of ultrasound therapy to a child and their family when presenting it as a recommended treatment option?

iii. Are the child and family involved in the decision to pursue ultrasound therapy?

iv. Are children more motivated to practice when they are using ultrasound?

v. Do you need to do frequent redirections of attention (to the ultrasound machine) during ultrasound therapy?

(1) If so, what percentage of kids need that?

(2) What would “frequent” look like?

vi. What do children and families get frustrated with about using ultrasound?

vii. What do children struggle with when using ultrasound?

viii. How do children constitute success? What does success look like to them?

Appendix B (p. 1 of 3)

Coding Manual

Showing codes in categories with each code's associated theme and frequency of appearance in the corpus. Theme abbreviations: EE = effort expectancy theme; FC = facilitating conditions theme; PE = performance expectancy theme; SI = social influence theme. Ultrasound is abbreviated here as "US."

Category: Barriers
Related to shortcomings or perceived negative features of ultrasound for speech sound therapy

- barrier.COVID – COVID-19 limited implementation of ultrasound (FC, 5)
- barrier.accessibility – limited availability/access to ultrasound and equipment (from client and clinician perspective) (FC, 15)
- barrier.age – limited ability to implement ultrasound due to client age (EE, 2)
- barrier.anatomy – anatomy interfering with image quality (e.g., cleft palate) (EE, 5)
- barrier.cost for client – expense of ultrasound (for institution and clients) (SI, 11)
- barrier.discomfort – related to client discomfort when holding ultrasound probe or gel (EE, 3)
- barrier.engagement – related to client boredom while using ultrasound (PE, 5)
- barrier.generalize – related to lack of generalization following intervention (PE, 7)
- barrier.imagequality – quality of image (poor) interfering with ultrasound utility (EE, 10)
- barrier.missinginfo – ultrasound is only able to capture the tongue; other factors might contribute to production (EE, 3)
- barrier.network – a limited network of collaborators can make implementation challenging (SI, 5)
- barrier.portable – limited ability to transport (FC, 3)
- barrier.setting – limited settings in which ultrasound is currently used/implemented (SI, 11)
- barrier.simpler – related to usability and intuitiveness of display (EE, 5)
- barrier.subjectivity – related to difficulty in determining clinical judgment of tongue position or accuracy of sound (EE, 4)
- barrier.training – limited graduate student training, continuing education (EE, 11)

Category: Benefit
Related to positive features of ultrasound for speech sound therapy

- benefit.engagement – client engagement improved with ultrasound (PE, 10)
    - benefit.curiosity – ultrasound facilitates client and parent interest (4)
    - benefit.motivation – how ultrasound can enhance motivation (2)
    - benefit.explore – related to how ultrasound allows for exploration of movement (4)
- benefit.complexity – ultrasound allows clinicians to appreciate complexity of speech (PE, 3)
- benefit.cues – ultrasound facilitates clinician cuing (with and without US) (PE, 7)
- benefit.dependency – unique to one clinician; they feel they cannot provide treatment without US anymore (PE, 1)
- benefit.ease – related to how ultrasound simplifies learning (EE, 5)
- benefit.education – related to how exposure to US on a student level (SLP students) can facilitate learning (PE, 2)
- benefit.familyinterpretation – how (easily) families are able to understand the image (PE, 7)
- benefit.motorcontrol – how ultrasound can illustrate progress at a motor control level (PE, 6)
- benefit.ownership – related to facilitating autonomy (SI, 1)
- benefit.professionalism – how US enhanced clinician professional development (PE, 4)
- benefit.progress – related to quicker or better progress with US compared to traditional US (PE, 7)
- benefit.realtime – real-time feedback facilitated progress (PE, 2)
- benefit.repetition – related to a specific question of how quickly clinicians can diagnose an articulatory pattern with US; clinicians identified that they can almost immediately see what's going on (PE, 3)

- benefit.treatingwithout – use of US facilitated how clinicians treat without US (PE, 7)
- benefit.visualization – being able to visualize the tongue was helpful for client, clinician, and family (PE, 15)

Category: Wants
Related to desired features, uses, or what ultrasound might accomplish

- wants.age – desire for ultrasound to relate to age of client (EE, 2)
- wants.2views – desire for multiple US views (e.g., coronal and sagittal) (EE, 2)
- wants.animation – desire for US to include an animated component or cartoon (EE, 3)
- wants.combination – desire for US to be combined with other component (e.g., MRI, tactile) (EE, 2)
- wants.draw – desire for drawing tool (EE, 2)
- wants.engagement – desire for US to be more engaging (EE, 1)
- wants.imagequality – desire for improvement in image quality (EE, 1)
- wants.portable – desire for increased portability (EE, 1)
- wants.simpler – desire for simpler display and controls (EE, 2)
- wants.vocaltract – desire to increase visualization to entire vocal tract (EE, 3)

Category: Progress
Related to factors that might limit or enhance progress

- progress.motivation – poor motivation might limit progress; good motivation might enhance progress (PE, 8)
- progress.evaluation – SLP uses clinical reasoning during evaluation to determine progress and potential for progress (PE, 2)
- progress.DD – developmental disabilities may limit progress (EE, 4)
- progress.ID – intellectual disabilities may limit progress (EE, 5)
- progress.age – age/maturity impact progress (EE, 5)
- progress.anatomy – differences in anatomy can influence image and subsequent changes (EE, 3)
- progress.apraxia – children with apraxia may show limited progress (EE, 2)
- progress.attention – attentional impairments (e.g., ADHD) may limit progress (EE, 3)
- progress.languagediversity – US may not be able to detect baseline for diverse speakers and adversely impact progress (SI, 1)
- progress.motorcontrol – limitations in fine motor control may limit progress (EE, 3)

Category: Experience
Related to topics covered in Part 1 of interview regarding clinician background. Largely coded for organization and easy referencing

- experience.numberclients – number of ultrasound clients seen by clinicians interviewed (EE, 8)
- experience.settings – setting in which clinicians used or saw ultrasound (might be current setting) (FC, 14)
- experience.supervision – clinician experience with supervising SLP students (EE, 10)
- experience.supervision.continueduse – follow-up question related to if supervised students have continued using ultrasound (FC, 4)
- experience.time – how long clinicians have used US (EE, 12)
- experience.yearstotal – total years of practice or experience without/before US (EE, 7)

Appendix B (p. 3 of 3)

Coding Manual

Category: Concerns
Related to family or child concerns and hesitancy that need to be addressed by clinicians before initiating treatment

- concerns.discomfort – related to physical discomfort (EE, 1)
- concerns.culture – related to cultural concerns using US (SI, 1)
- concerns.germs – related to infection (EE, 1)
- concerns.heat – related to risk of heat (EE, 2)
- concerns.medicaldevice – related to anxiety using a medical device (EE, 3)
- concerns.radiation – related to allaying concerns of (nonexistent) ionizing radiation from others (EE, 2)

Category: Success
Related to how children constitute success

- success.feedback – rely on feedback from clinicians or parents (SI, 6)
- success.meaningfulwords – able to say important words (e.g., name, sibling name) (SI, 1)
- success.social – related to social aspect of speech (e.g., less bullying, school participation) (SI, 4)
- success.understood – related to intelligibility of speech (SI, 5)

Miscellaneous Category
Related to answers given in response to specific content-focused questions asked by the interviewer

- bias – related to specific question about bias in "correctness" and suggestions to minimize bias (SI, 13)
- image.identification – related to question of if clinicians can consistently identify what's going on in image (EE, 6)
- perception – related to question about if clinicians rely on perception, image, or combination (PE, 13)
- qualifications – related to question of qualifications for clinicians using US (EE, 29)
- variety – specific quotes about how a variety of tongue shapes and individual differences can change treatment and visualization (EE, 9)